Strain analysis: condition, check and damage ranking

Critical strains from the backcalculated moduli (26,048 usable bowls, 50 kN wheel, 800 kPa, verified against ERAPave within 0.34%) · Östergötland 2022–2024

Summary

- **Condition:** with the Finnish strain limits, 22% of the measured road is alarming or deteriorating in Sep 2024 (4% with the Swedish deflection formula). The share depends on the survey and on which strain is used, so the absolute class is uncertain.
- **Check against Trafikverket:** our strain and the official Swedish formula (Eq. 3) rank the road almost the same way (r = 0.91), but ours is about 30% higher. 97% of the points fall in the same class or a neighbouring one.
- **Damage ranking:** 299 road segments of 500 m were ranked by damage per year (strain^4 × heavy traffic). The 10% worst segments carry 34% of all fatigue damage. Roads 32 and 50 dominate the top. The ranking hardly changes without the bowls at a modulus limit (Spearman 0.96 / 0.97).
- **SCI200 and BCI** with the Finnish limits do not separate good from bad roads here: SCI200 puts more than 99% of the network in one class, BCI 67–92%.

1. Condition classes (Finnish limits, Saarenketo & Männistö 2024)

Each usable 20 m point was classified with the limits from Table 1 of the printed paper: asphalt strain <100 good, 100–200 adequate, 200–300 satisfactory, 300–400 alarming, >400 deteriorating με; SCI200 = D0 − D200 and BCI = D900 − D1200 with their own limits. The bars show the share of the measured road per class.

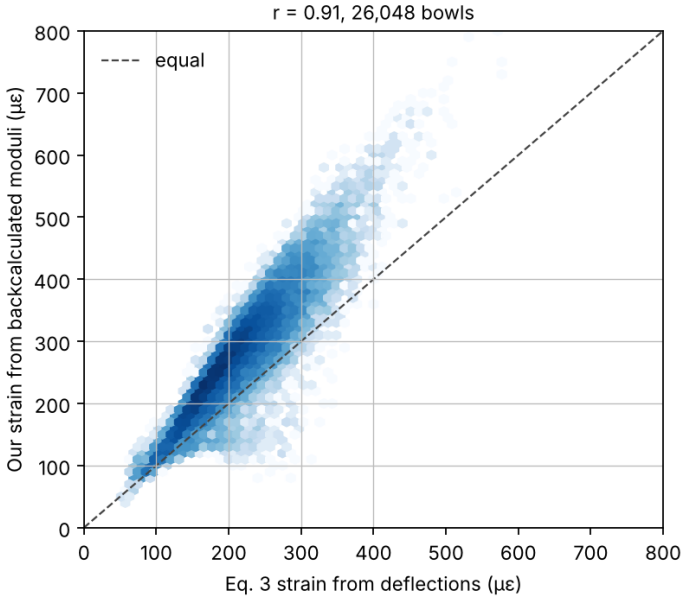


Alarming + deteriorating	Aug 2022	Sep 2023	May 2024	Sep 2024
Asphalt strain, ours	44%	38%	44%	22%
Asphalt strain, Eq. 3	12%	7%	13%	4%

- The class depends on the survey: Sep 2024 looks clearly better. The moduli are corrected to 20 °C, so this is not temperature; part of it is the season and the trade-off between layers in the backcalculation. A fair network condition should therefore be based on several surveys.
- SCI200: the TSD values are low (median 106 µm, 95% below 183 µm), far below the first Finnish limit of 250 µm. The Finnish limits were made for thin Finnish pavements and are not directly transferable to TSD (Hamidi et al. 2024; Katicha et al. 2014).

2. Our strain vs the Swedish deflection formula (Eq. 3)

Eq. 3 (TRVMB 114, Trafikverket 2012:051, quoted in SBUF 13019): ε = 37.4 + 0.988·D0 − 0.553·D200 − 0.502·D600, for FWD at 50 kN. TSD has no 200 mm sensor, so D200 was interpolated between D130 and D215. Both strains use the same temperature-corrected bowls.



	Value
Median strain, ours / Eq. 3	269 / 207 με
Correlation (Pearson / Spearman)	0.91 / 0.90
Without bowls at a modulus limit	r = 0.91
Relation	ours ≈ 1.35 × Eq. 3 − 12 με
Same class / same or neighbouring class	40% / 97%

- **Same ranking:** both methods agree on which stretches have high and low strain. This is an independent check that the backcalculated moduli are realistic.
- **Different level:** our strain is about 30% higher. Likely reasons: Eq. 3 is an empirical formula fitted to FWD data, TSD bowls are not the same as FWD bowls, and D200 is interpolated. Because the class limits are fixed numbers, this level difference moves many points one class up.

3. Damage ranking (where the road wears out fastest)

Following Saliko & Erlingsson (2021): allowed passes $N = f \cdot \varepsilon^{-4}$, so the damage of one truck is proportional to ε^4 (Miner's rule). Damage per year = heavy vehicles/day × 365 × (ε / network median)^4, averaged over the usable surveys of each location and over 500 m segments. Index 1 = a segment with typical traffic (133 heavy vehicles/day) and typical strain. Fatigue uses the asphalt tension, rutting the subgrade compression. The unknown constant f cancels out, so the index ranks segments but does not give years of life. 35 segments where most bowls have a modulus at a limit are left out of the lists.

Top 10 segments, fatigue cracking

Rank	Road	km	Heavy veh./day	Asphalt (mm)	Strain (με)	Class	Fatigue index
1	50	3.5–4.0	709	180	358	Alarming	20.4
2	50	3.0–3.5	709	176	335	Alarming	20.0
3	50	14.0–14.5	674	135	363	Alarming	17.3
4	50	13.5–14.0	674	136	337	Alarming	16.4
5	50	2.5–3.0	709	186	320	Alarming	14.2
6	32	6.0–6.2	568	162	337	Alarming	14.1
7	50	13.0–13.5	674	123	299	Satisfactory	10.9
8	32	15.0–15.5	650	146	307	Alarming	10.0
9	1136	3.0–3.5	539	126	316	Alarming	9.4
10	50	2.0–2.5	709	173	277	Satisfactory	9.1

Top 10 segments, rutting (subgrade)

Rank	Road	km	Heavy veh./day	Subgrade strain (με)	Rutting index
3	50	13.0–13.5	674	110	195
4	1050	5.0–5.5	285	129	188
5	1050	22.0–22.5	193	137	145
6	50	6.2–6.4	640	104	142
7	50	13.5–14.0	674	90	133
8	50	15.3–15.5	674	82	112
9	1050	4.1–4.5	285	120	111
10	741	32.0–32.5	59	177	98
11	1136	2.5–3.0	539	55	94
12	50	5.0–5.3	709	100	91

Ranks are among all 299 segments; missing ranks belong to segments flagged as uncertain. Full list: damage_ranking_500m.csv.

- **Traffic matters as much as strain:** roads 50 and 32 carry 570–710 heavy vehicles/day, so even satisfactory strain gives a high index. Road 4 (2,163 heavy vehicles/day) enters the top list with only satisfactory strain.
- **Damage is concentrated:** 10% of the segments carry 34% of all fatigue damage, which supports targeted maintenance.
- **Rutting is more uncertain:** subgrade strain changes strongly with season (the wet Sep 2023 survey doubled it), and the power 4 enlarges single high values.

Limitations

- Linear elastic model, standard load (50 kN single wheel), no temperature variation over the year; bottom-up cracking only (no top-down cracking).
- Eq. 3 and the Finnish limits were developed for FWD; TSD values may need calibration before absolute classes are used.
- The damage index ranks segments; remaining life in years needs the coefficients f (TRVK Väg) and the age of each road.

Code: strain/calc_strains_all.py and strain/strain_analysis.py. Outputs in data/strain_analysis/.